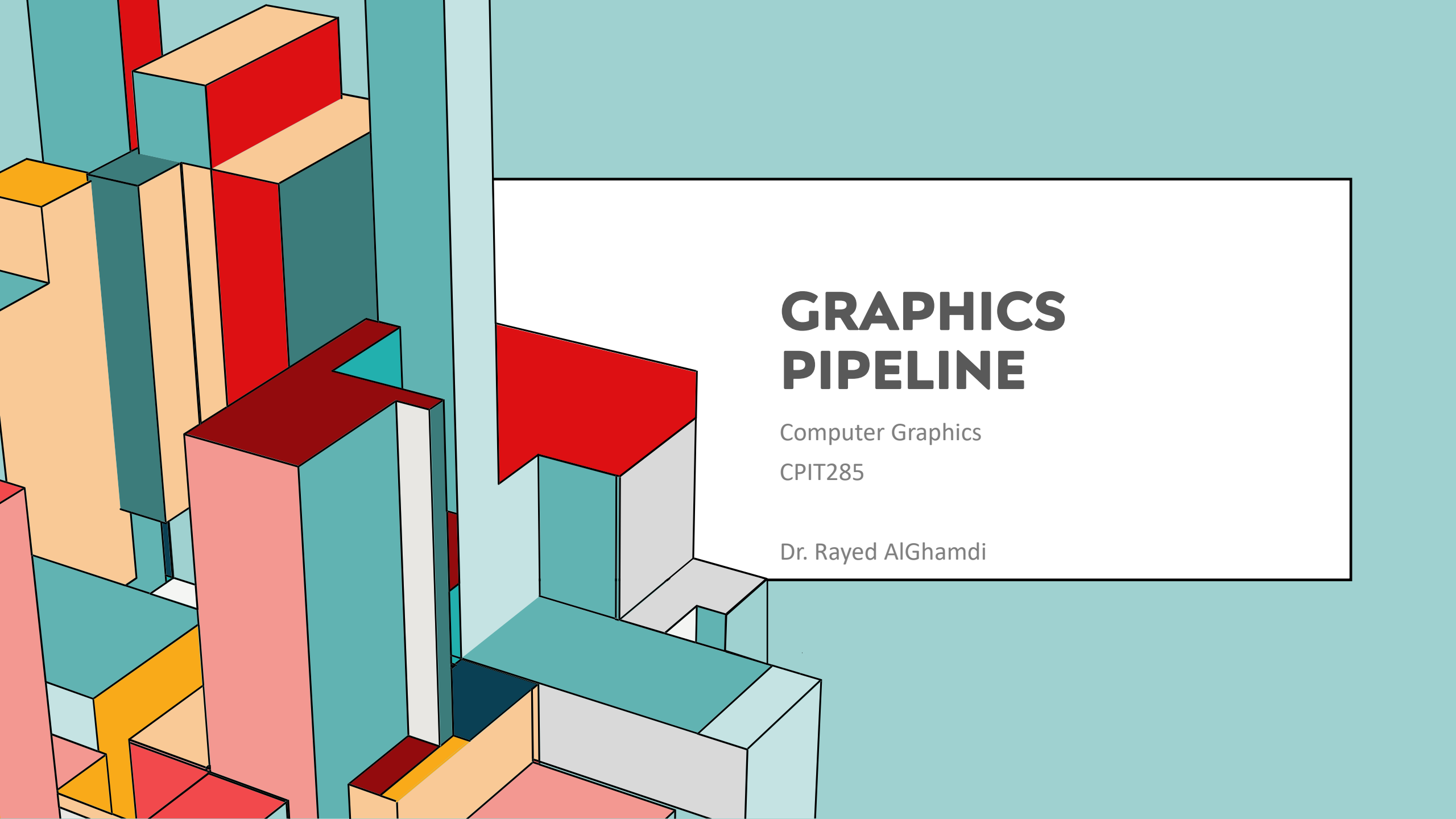




GRAPHICS PIPELINE

Computer Graphics

CPIT285

Dr. Rayed AlGhamdi

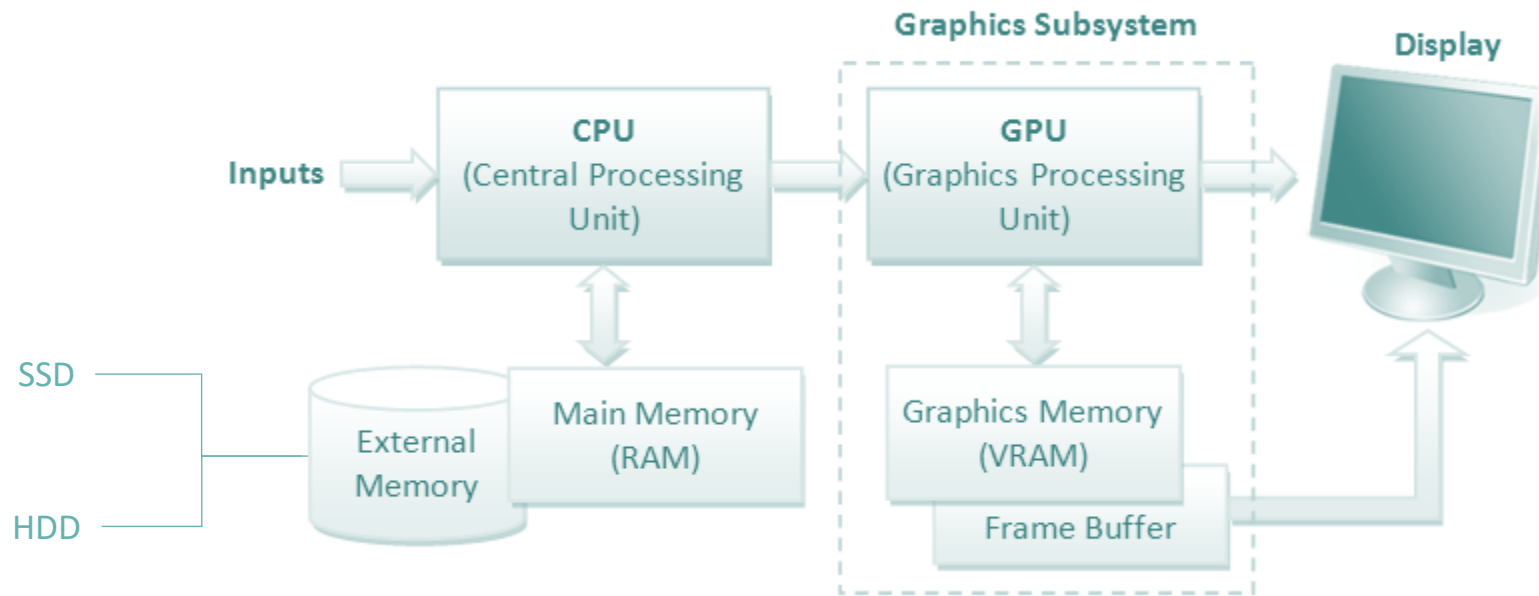
GRAPHICS PIPELINE

- A computer **graphics pipeline**, which can be called rendering pipeline, is a model that describes the needed steps to display a 3D scene to a 2D screen.
- Because the steps required for this operation depend on the software and hardware used, there is no universal graphics pipeline suitable for all cases.
- However, graphics application programming interfaces (**APIs**) such as OpenGL and Direct3D were created to unify similar steps.
- These APIs abstract the underlying hardware and keep the programmer away from writing code to manipulate the graphics hardware (NVIDIA, Intel, AMD, etc.).

GRAPHICS HARDWARE

GPU

- Modern computer has dedicated **Graphics Processing Units (GPU)** to produce images for the display, with its own graphics memory (or Video RAM or VRAM).



GRAPHICS HARDWARE

Pixels and Frame

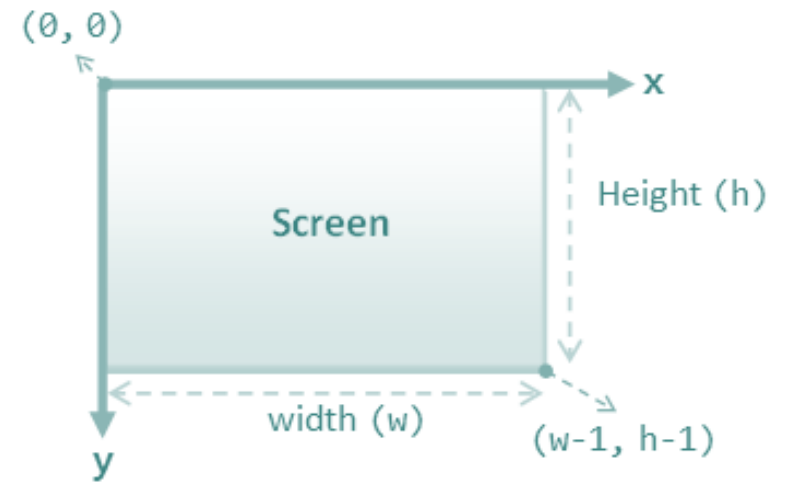
- Monitor displays are raster-based. A **raster** is a 2D rectangular grid of pixels.
- A pixel has two properties: a color and a position.
- A common use for coloring is 8 bits representing the value of each of the three main colors RGB: Red, Green & Blue ($8 \times 3 = 24$ bits)
- The number of color-bits per pixel is called the **color depth** (or precision) of the display.



GRAPHICS HARDWARE

Pixels and Frame

- The position is expressed in terms of (x, y) coordinates. The origin $(0, 0)$ is located at the top-left corner, with the X-axis pointing right and the Y-axis pointing down.
- The number of pixels (rows by columns) of the rectangular grid is called the **display resolution**, which can range from 640x480 (VGA), 800x600 (SVGA), 1280x720 (HD) to 1920x1080 (FHD), or even higher.

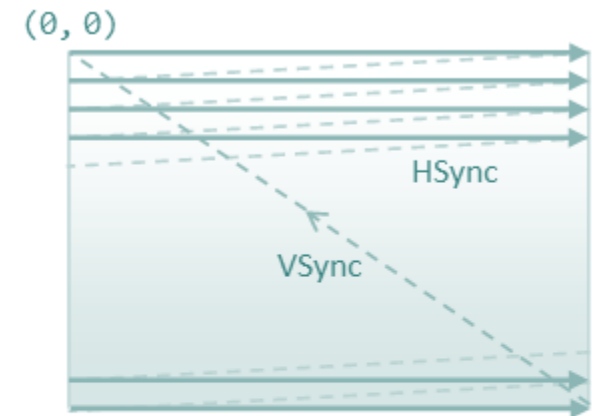


The 2D Screen Coordinates: The origin is located at the top-left corner, with x-axis pointing left and y-axis pointing down.

GRAPHICS HARDWARE

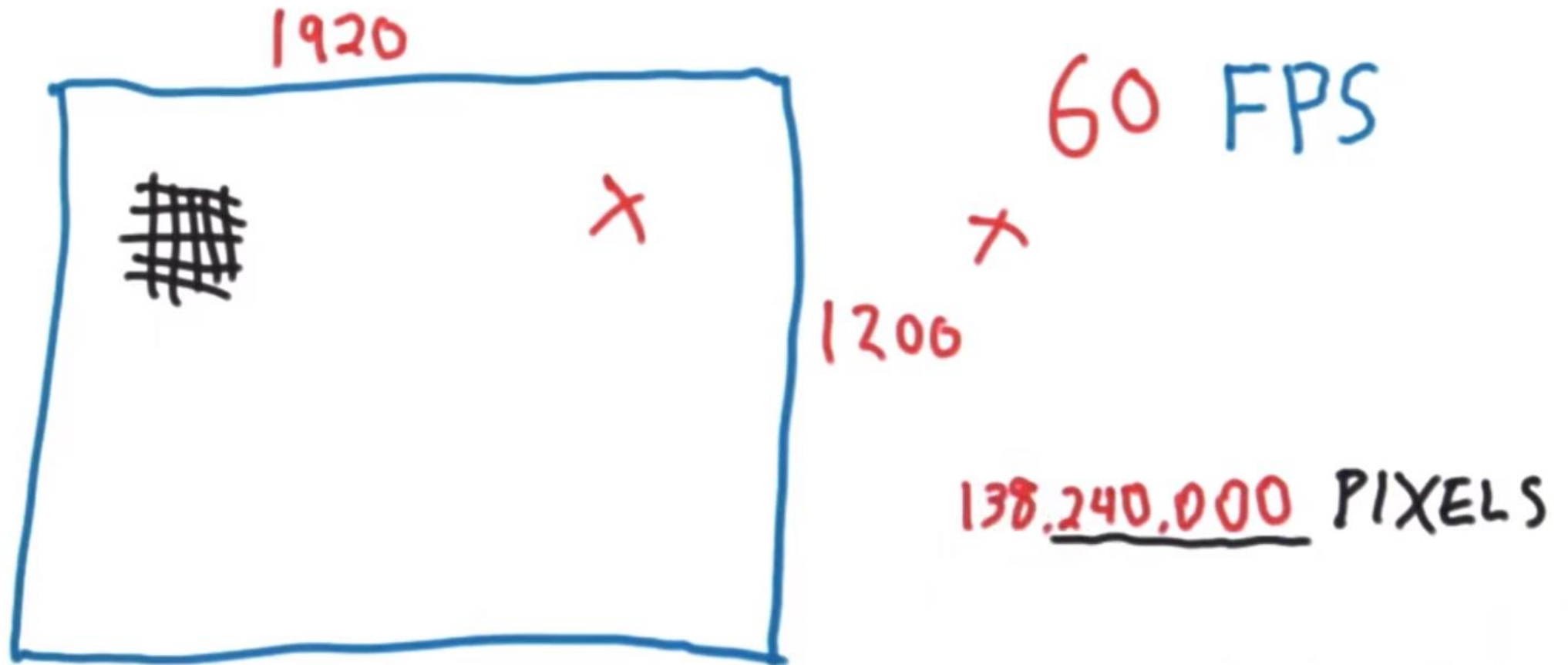
Frame Buffer and Refresh Rate

- The color values of the pixels are stored in a special part of graphics memory called the **frame buffer**.
- The GPU writes the color value into the frame buffer. The display reads the color values from the frame buffer row-by-row, from left-to-right, top-to-bottom, and puts each of the values onto the screen. This is known as **raster-scan**.
- The display refreshes its screen several dozen times per second, typically 60Hz. This is known as the **refresh rate**.
- A complete screen image is called a frame. The term **FPS** is to measure how many frames are presented per second.



Raster Scan: The display updates its contents row-by-row from top-to-bottom, left-to-right by reading the color value of the pixels from the frame buffer.

How many pixels must be computed each second to attain the rate of 60 FPS on a 1920 x 1200 screen?

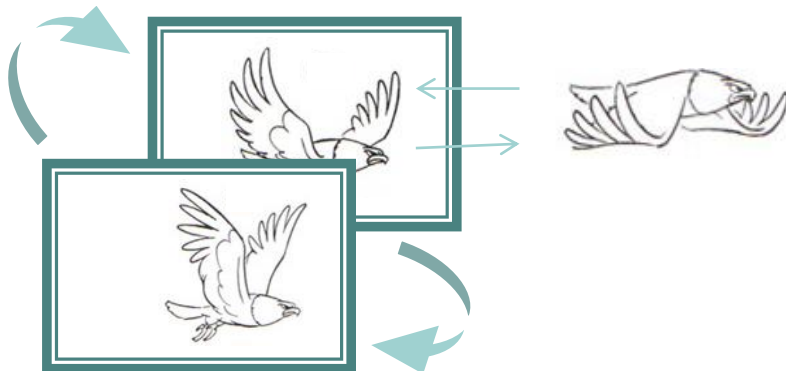


GRAPHICS HARDWARE



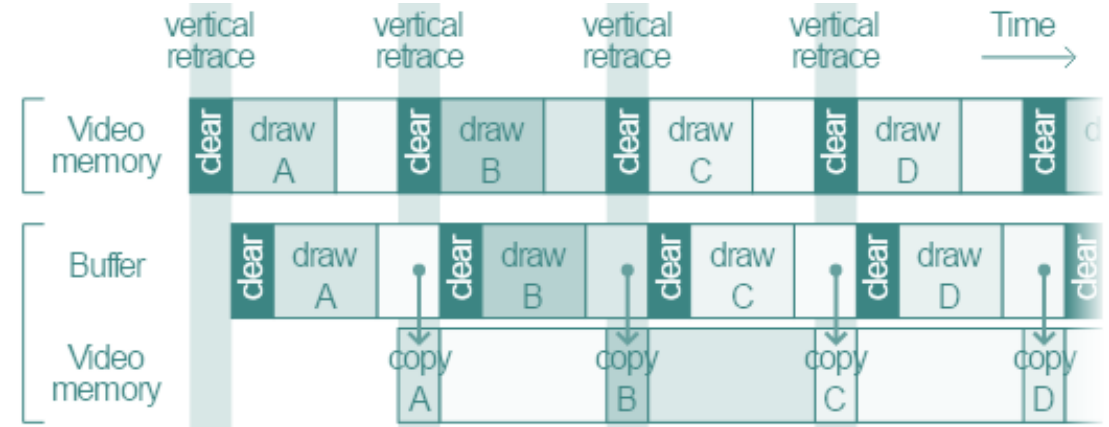
Double Buffering

- Modern GPU uses **double buffering**: front and back. The display reads from the front buffer, while we can write the next frame to the back buffer. When we finish, we signal to GPU to swap the front and back buffer (known as buffer swap or page flip).
- This is important in animated applications
- Have you heard of '**Triple Buffering**'?



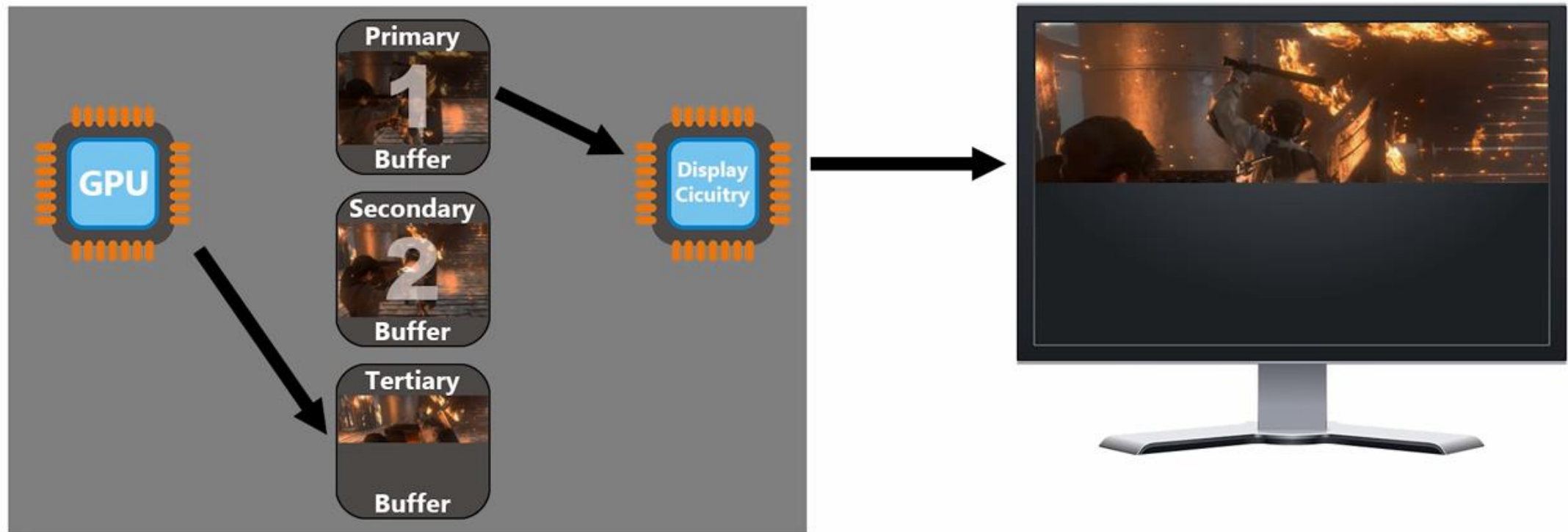
1. Single-buffering

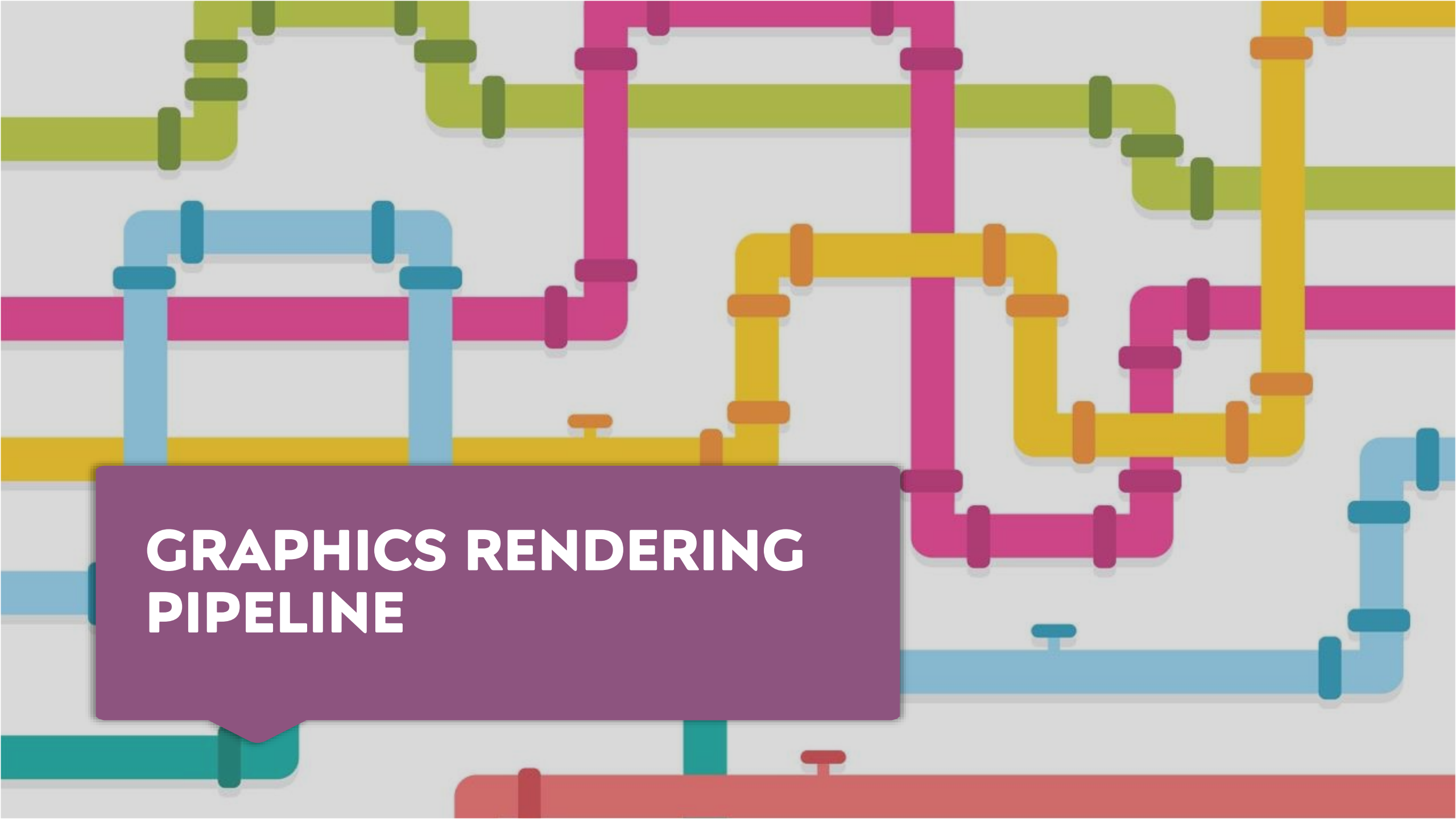
2. Double-buffering



GRAPHICS HARDWARE

Triple Buffering





GRAPHICS RENDERING PIPELINE

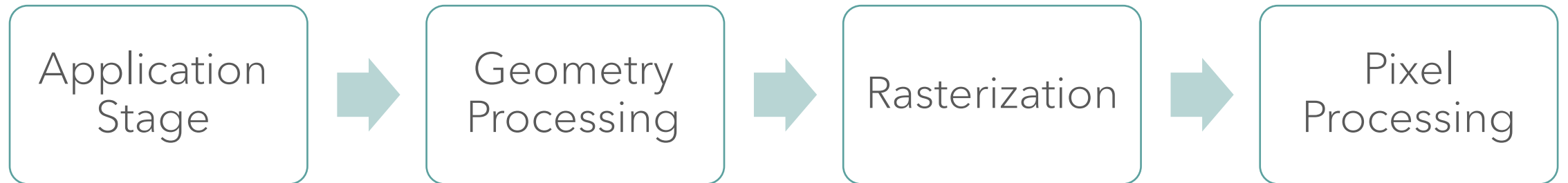
GRAPHICS RENDERING PIPELINE

In computing terminology...

- A pipeline refers to a series of processing stages in which the output from one stage is fed as the input of the next stage, similar to a factory assembly line or water/oil pipe.
- In computer graphics, rendering is the process of producing an image on the display from a model description.
- The 3D Graphics Rendering Pipeline accepts the description of 3D objects in terms of vertices of primitives (such as triangle, point, line, and quad), and produces the color-value for the pixels on the display.

GRAPHICS RENDERING PIPELINE

The Processing Stages



GRAPHICS RENDERING PIPELINE

The Processing Stages

- The 3D graphics rendering pipeline consists of the following main stages:
 - **Application Stage**: initializing the window where rendered graphics will be displayed; sending data to the GPU.
 - **Geometry Processing**: determining the position of each vertex of the geometric shapes to be rendered, implemented by a program called a vertex shader.
 - **Rasterization**: determining which pixels correspond to the geometric shapes to be rendered.
 - **Pixel Processing**: determining the color of each pixel in the rendered image, involving a program called a fragment shader.

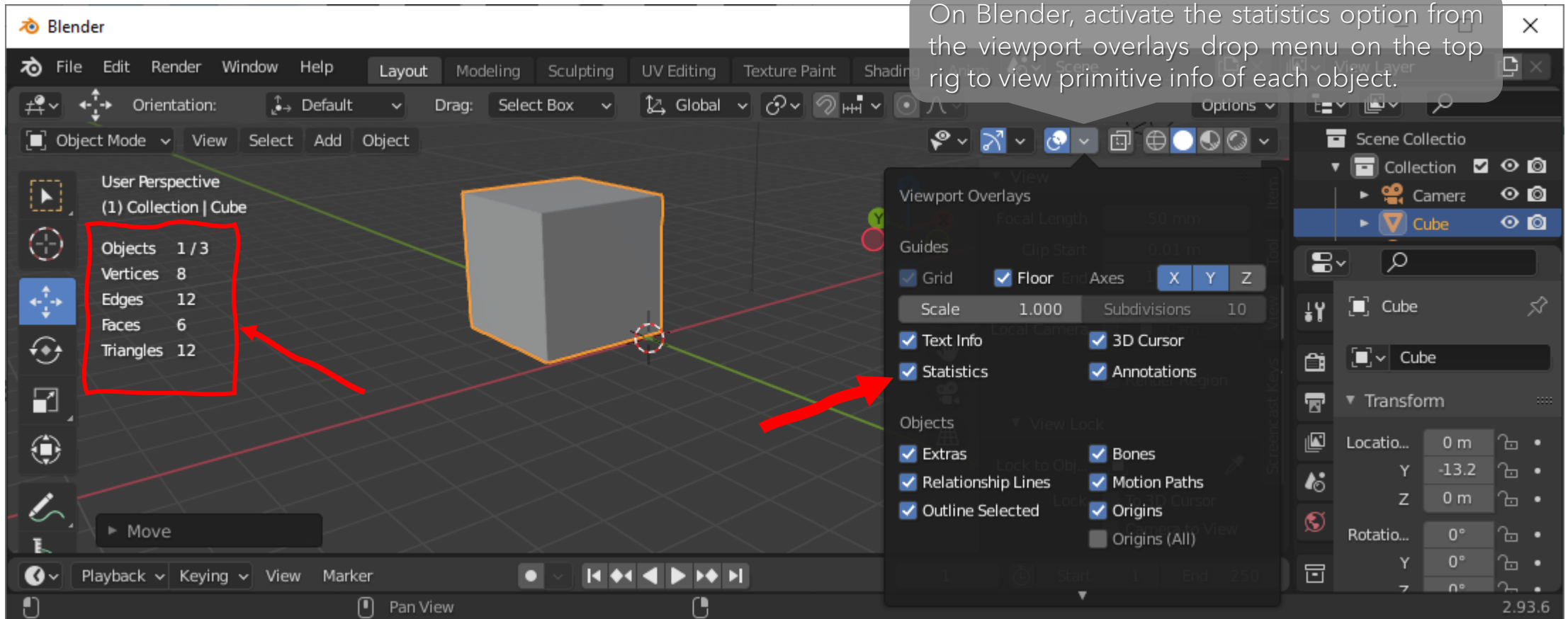
VERTICES, PRIMITIVES, FRAGMENT & PIXELS

Vertices & Primitives

- The inputs to the Graphics Pipeline are geometric primitives.
- A primitive is made of one or more vertices such as points, lines, and closed polygons (triangles, quads,...).
- A vertex is a point in the coordinate system.
- A coordinate system is made of 2 axes (x & y) in 2D graphics and 3 axes (x, y & z) in 3D graphics.
- Each vertex is associated with its attributes: position, color, normal and texture.
- Cube, Sphere, and pyramid are not primitives. They are typically assembled using primitive triangles or quads.

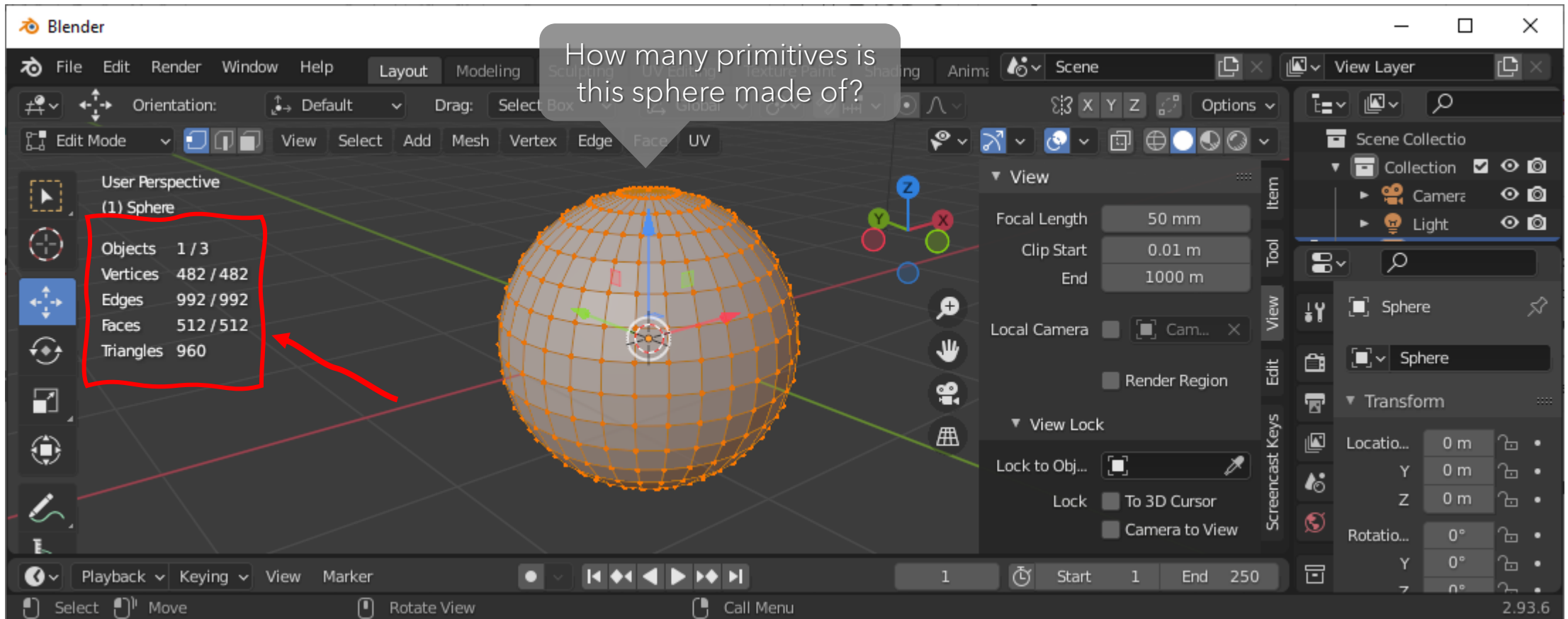
VERTICES, PRIMITIVES, FRAGMENT & PIXELS

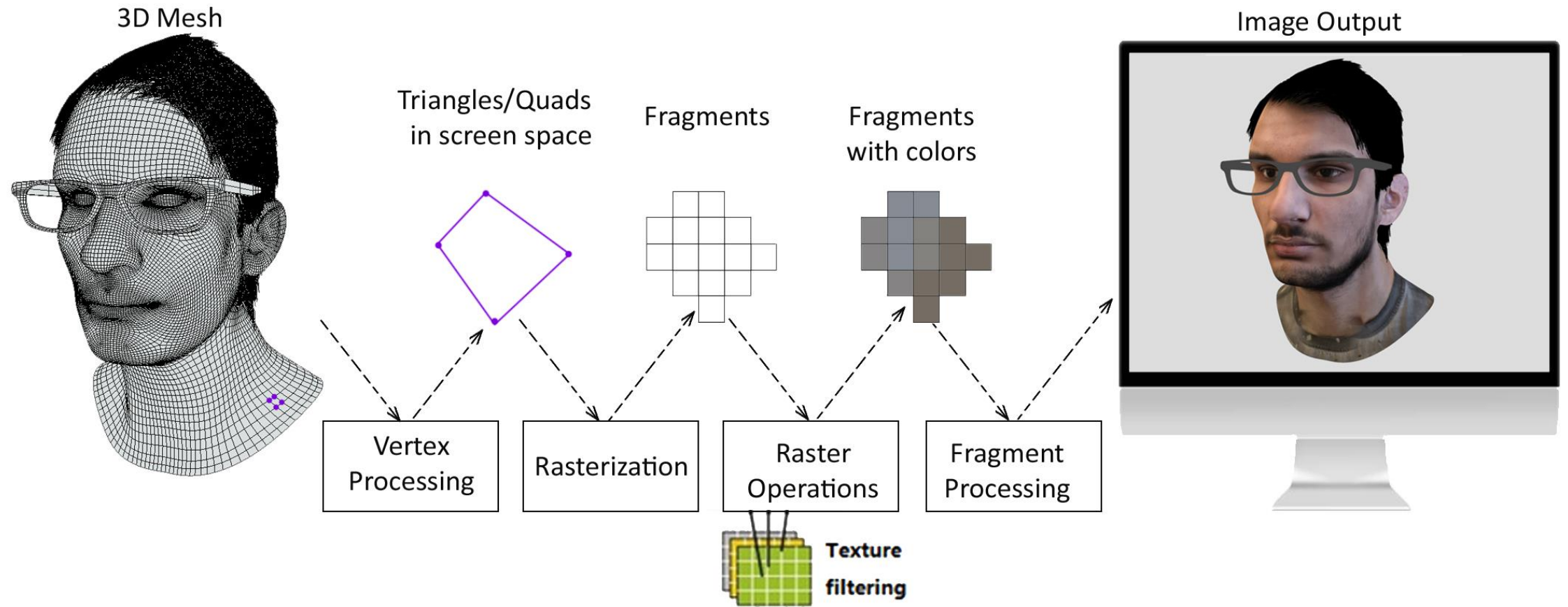
Vertices & Primitives



VERTICES, PRIMITIVES, FRAGMENT & PIXELS

Vertices & Primitives

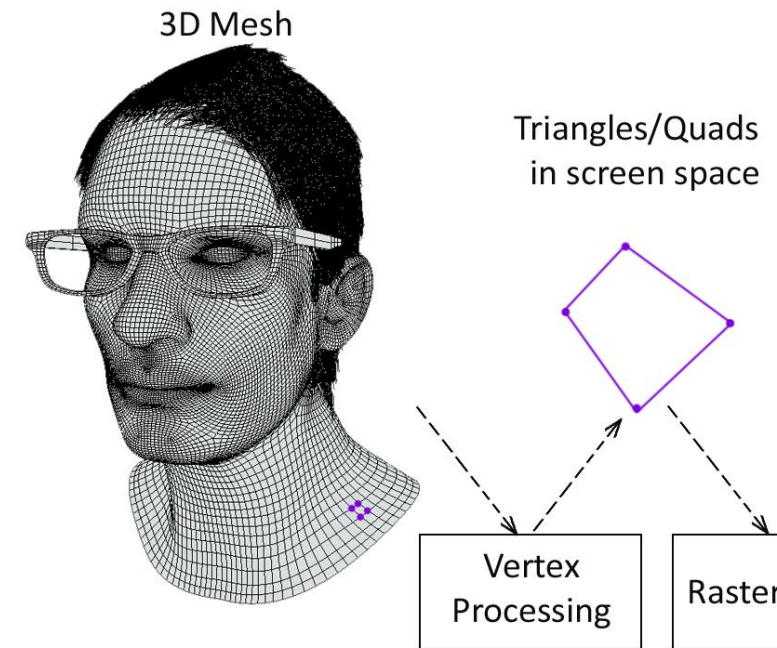




VERTICES, PRIMITIVES, FRAGMENT & PIXELS

Vertex Processing

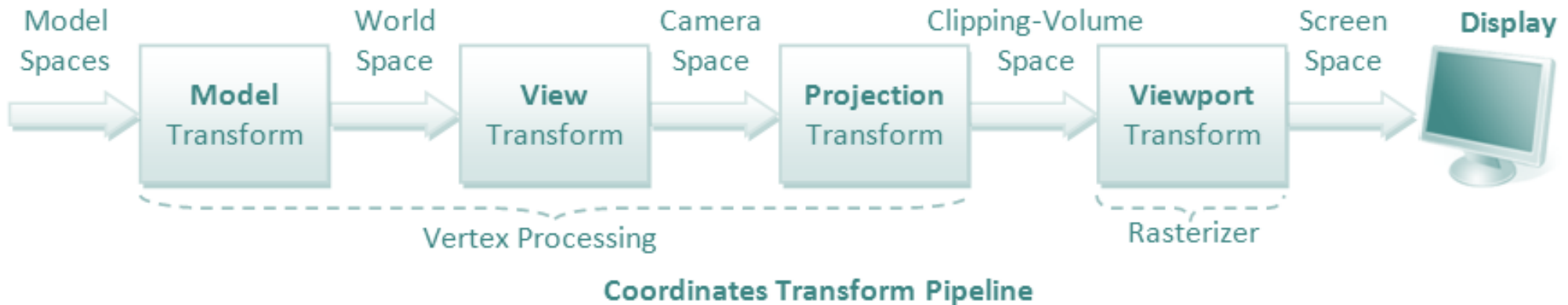
- The process used to produce a 3D scene on the display in Computer Graphics is like taking a photograph with a camera.
- It involves four transformations:
 1. Arrange the objects in the world (**Modeling Transformation**).
 2. Position and orientation of the camera (**Viewing Transformation**).
 3. Select a camera lens (wide-angle, normal, or telescopic), adjust the focal length and zoom factor to set the camera's field of view (**Projection Transformation**).
 4. Print the photo on a selected area of the paper (**Viewport Transformation**) – in the rasterization stage.



VERTICES, PRIMITIVES, FRAGMENT & PIXELS

Vertex Processing

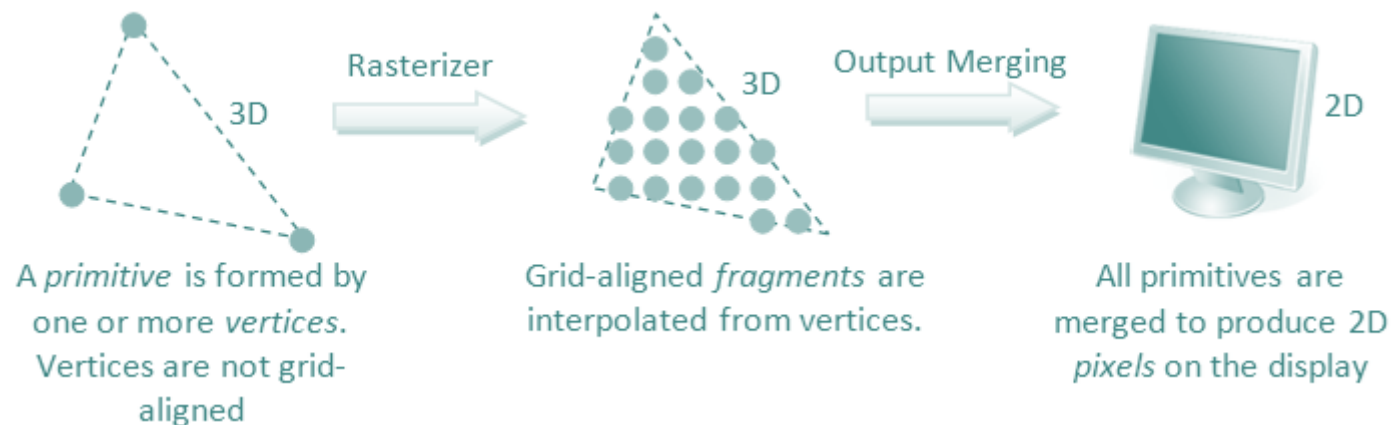
- The four transformations steps are further detailed in chapters 6 and 7



VERTICES, PRIMITIVES, FRAGMENT & PIXELS

Pixel vs. Fragment

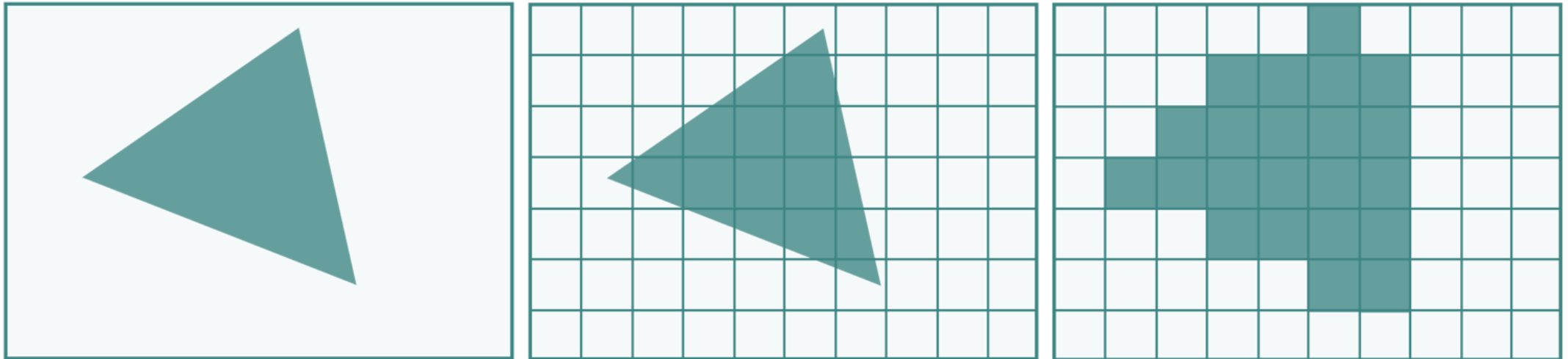
- Pixels refer to the dots on the display screen, which are aligned in a 2-dimensional grid of certain rows and columns corresponding to the display's resolution.
- A pixel is 2-dimensional, with (x, y) position and RGB color value.
- The purpose of the Graphics Rendering Pipeline is to produce the color-value for all the pixels for displaying on the screen, given the input primitives.



VERTICES, PRIMITIVES, FRAGMENT & PIXELS

Pixel vs. Fragment

- In order to produce the grid-aligned pixels for the display, the rasterizer of the pipeline, takes each input primitive and performs a raster-scan to produce a set of grid-aligned fragments enclosed within the primitive.



VERTICES, PRIMITIVES, FRAGMENT & PIXELS

Pixel vs. Fragment

- A fragment is 3-dimensional, with a (x, y, z) position.
- The (x, y) are aligned with the 2D pixel grid. The z -value (not grid-aligned) denotes its depth. The z -values are needed to capture the relative depth of various primitives so that the occluded objects can be discarded in the output-merging stage.
- Fragments are produced via interpolation of the vertices. Hence, a fragment has all the vertex's attributes such as color, fragment-normal and texture coordinates.
- In modern GPU, vertex processing and fragment processing are programmable. The programs are called vertex shader and fragment shader.

 NVIDIA®
GEFORCE
RTX



**What is RTX stand
for?**

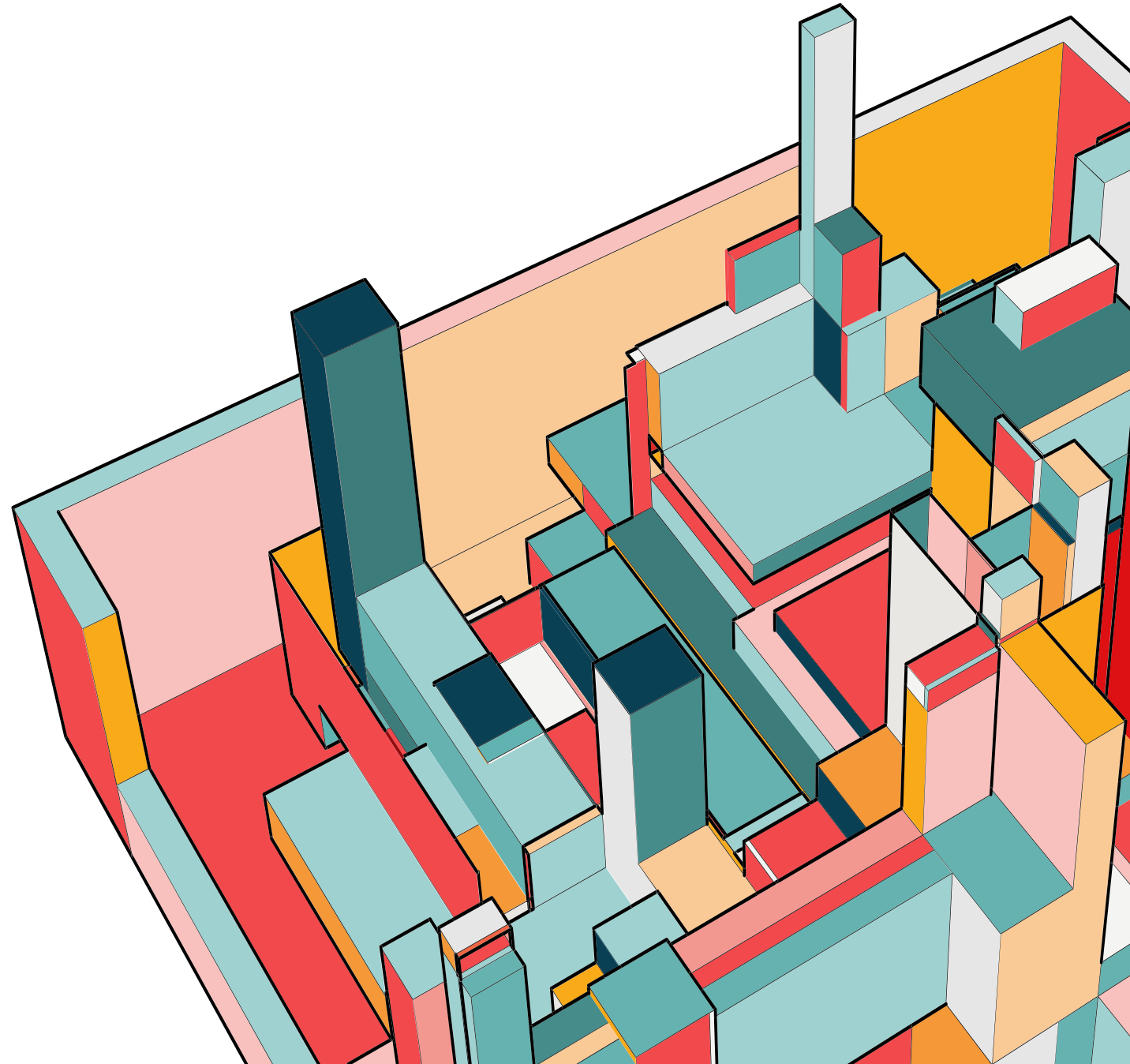
Ray Tracing texel eXtreme

- How does it work?
- Is it the replaceable technique of rasterization?

100% CONFIDENT

Your journey in this course is
enjoyable and beneficial!

Enjoy learning CPIT285!



THANK YOU

Dr. Rayed ALGhamdi

+966-12-640-0000 ex:75560

raalghamdi8@kau.edu.sa

www.raalghamdi8.kau.edu.sa

